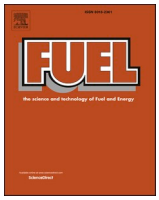
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Full Length Article



Promoting the production of methane on the co-digestion of food waste and sewage sludge by aerobic pre-treatment

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ABSTRACT

In this study, methane (CH_4) production performance on the co-digestion of food waste (FW) and sewage sludge (SS) was compared under non-pretreatment and 6 days aerobic pre-treatment. The results showed that, compared with the non-aerobic group, the CH_4 productivity increased by nearly 60% through aerobic pre-treatment. More ammonium was obtained after pre-aeration, providing a suitable growth environment for methanogens. After the aerobic pretreatment, proteins and carbohydrates converted into solute protein and carbohydrate, which could be easily consumed by microorganisms, concentrations of solute protein and carbohydrate were 350 and 667 mg/L in R2 and Pre-R2, respectively. At the same time, aerobic pretreatment affected the distribution of microbial community in the system. In the stable stage of Pre-R2, *Methanosarcina* as acetate-utilizing methanogen, became the dominant in the reactor, whose abundance was over 93%. It meant after the aerobic pretreatment, the main pathway for methanogenesis was acetotrophy. This study revealed that pre-aeration is a simple and desirable strategy for improving the CH_4 productivity on the co-digestion of FW and SS.

1. Introduction

As the main part of municipal solid waste (MSW), the disposal of FW has become an urgent issue. The amount of FW is increasing year by year. It is reported that 1.3 billion tons of FW has been generated every year all over the world [1]. Particularly in China, the rapid increase in FW is triggered by population growth and urbanization. More than 80–110 million tons of FW is produced annually, accounting for 40–60% of MSW in China [2]. FW contains high content of organic matter, which is easily degraded in a short period of time, thus producing serious odors. This also results in breeding mosquitoes and flies, thereby, destroying the environment and spreading diseases. Once the leakage occurred during the process of FW collection and transportation, it would pose threat to the human health and environment [3,4]. The traditional technologies for the treatment of food waste are composting, incineration and sanitary landfill, but, they are proved to, not only cause the loss of chemical values of FW, but also result in secondary pollution, like the production of dioxins and 2,3,7,8-tetrachlorodibenzo-p-dioxin (2,3,7,8-TCDD), these by-products are toxic to human health [5,6]. Anaerobic digestion refers to the process of decomposing biodegradable organic matter into biogas by anaerobic microorganism. It is an attractive technology and holds dual-advantage, i.e., reducing organic waste and resulting in energy recovery [7].

Due to the abundance of protein, lipids and carbohydrates, food waste has been considered to be a potential substrate in the anaerobic digestion process for the degradation and production of CH_4 [8]. Numerous researchers have carried out the anaerobic digestion of food waste and found that because of the VFAs accumulation, imbalance of C/N ratio, insufficient trace elements in the anaerobic digestion process of the food waste, low CH_4 yield was obtained [4]. It has been reported that when the organic loading rate (OLR) is higher than 2.5 g VS/L/d under thermophilic conditions, single anaerobic digestion of food waste failed due to the accumulation of VFAs and ammonia [9]. In order to improve the performance of the anaerobic digestion system, an array of strategies are proposed, such as adding trace elements [10], pre-treatment methods (including sonication pretreatment [11], ionizing radiation [12]; mechanical grinding [13]). Although anaerobic digestion performance of food waste can be enhanced by these methods, but, most of these methods are costly and it is difficult to apply them commercially into the plants. As a popular strategy, co-digestion can

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increase the CH_4 production yield with following advantage: (1) supplying additional nutrients; (2) allowing for treatment of different organic wastes in a single facility; (3) refining the buffer capacity and (4) improving the stability of the anaerobic digestion system [14,15]. Because of the abundance of microorganisms and micronutrients, SS is commonly used as co-substrate for FW [9].

Compared with the chemical and physical pre-treatment, aerobic pretreatment, as a relatively cheap and easy pre-treatment technology (only need to pump oxygen into the system), is favored as an appropriate option to stimulate CH_4 production from MSW, lignocellulosic biomass and SS [16,17]. Ni et al. [18] studied the effect of pre-aeration time on CH_4 and hydrogen production from municipal solid waste and concluded that aerobic pretreatment can significantly improve the performance of dry anaerobic digestion, that is, an optimum pre-aeration reduced the organic loading and enhanced the buffer capacity of the reactors. The effect of aerobic pretreatment on the CH_4 production of food waste from different materials (protein, lipids and carbohydrate) has also been studied by Rafieenia [19], at a two-stage anaerobic digestion process, they found the effect of aerobic pretreatment is greatly related to different feedstock, pre-aeration showed positive effects on CH_4 production for protein rich and carbohydrate rich samples. González-González et al. [20] has showed that aerobic pre-treatment is profitable for olive oil anaerobic digestion, after 5 days, the CH_4 yield was 2.4 times higher than untreated and 1.4 times higher than 7 days aerated treatment. The aerobic pre-treatment can degrade the organics of substrates, thus avoiding the overload of organic and promoting the CH_4 production. However, there has been no studies focused on the same OLR during the digestion process and pre-aerated and non-pretreated substrates were introduced as feed in the subsequent anaerobic process. In order to reveal the effect of pre-aeration on the substrates, in this research, the natural reoxygenation pretreatment on the substrate are investigated in a semi-continuous system without adding any additional bacteria. The effects of aerobic pre-treatment on the subsequent methanogenesis are also investigated with the pretreated and untreated substrate as the raw feed material (the same high loading rate in anaerobic digestion stage, OLR = 4 g VS/L/day). In addition, to ensure the stability of the anaerobic digestion system under high OLR, co-digestion of FW and SS with different ratios are also considered. To find out the difference of substrates treated with and without the pre-aeration, the fraction of substrates was analyzed. Meanwhile, the microbial distribution at the final running stage of anaerobic digestion was measured.

2. Materials and methods

2.1. FW and SS

FW used in this study was collected from the school canteen of Xi'an Jiaotong university. Undesirable materials (like tissue, bones and plastics) were manually screened out. In order to get as many diverse ingredients as possible, the food waste samples were taken from different waste disposal buckets and mixed thoroughly. They were then crushed into viscous liquid and stored in a −20 °C refrigerator before carrying out the experiments. SS came from one of the wastewater treatment plants in Xi'an, China. The detailed information about food waste and sewage sludge is summarized in Table 1.

2.2. Inoculum

The inoculum was obtained from mesophilic anaerobic digester of a municipal sludge treatment plant in Dalian, China. The pH of inoculum was 7.45 ± 0.12 and total solid (TS) was 68.5 ± 1.6 g/L with a 46.1% VS/TS ratio. Prior to the experiment, the inoculum was degassed for 5 min.

Table 1
Characteristics of food waste and sewage sludge.

Parameters	Unit	Value	
		FW	SS
pH	–	6.5 ± 0.1	8.1 ± 0.1
TS	g/L	225 ± 0.5	215 ± 0.8
VS	g/L	210 ± 2.0	112.5 ± 1.3
VS/TS	%	93.33	53.33
Protein	g/L	39.8 ± 1.1	76.92 ± 2.0
Lipid	g/L	73.04 ± 3.5	16.72 ± 0.8
C	% TS	49.52 ± 0.1	20.88 ± 0.1
H	% TS	7.47 ± 0.1	3.37 ± 0.1
N	% TS	1.90 ± 0.1	3.57 ± 0.1
Ash	% TS	6.67 ± 0.1	47.67 ± 0.1
C/N	–	26.11	5.84
Ammonia	g/L	0.20 ± 0.04	1.09 ± 0.01
Ca	g/kg TS	1.27 ± 0.1	34.50 ± 2.3
Mg	g/kg TS	0.35 ± 0.3	4.45 ± 0.5
Al	g/kg TS	0.19 ± 0.1	26.2 ± 2.1
Fe	g/kg TS	0.51 ± 0.01	28.5 ± 3.0
Zn	g/kg TS	0.06 ± 0.02	0.62 ± 0.1
Mn	g/kg TS	0.03 ± 0.01	0.44 ± 0.2

2.3. Experimental procedure

The detailed steps of experiment are designed as follows.

(1) A 6 days natural pre-aeration with different FW:SS ratio of substrates.
(2) The substrate treated by pre-aeration and original substrate were used as the feed for the anaerobic digestion process under the same OLR.

2.3.1. Aerobic pre-treatment of substrates

To compare the substrates with and without aerobic pre-treatment, two arrays were performed. Moreover, experiments were also performed with different FW:SS ratio. The pre-treatment array was carried out as follows: substrates were prepared at designated ratios (FW(VS):SS (VS) = 1:0, 3:1, 1:1, VS calculated). Then, the obtained substrates were further diluted to 500 mL (with distilled water) in three identical beakers, respectively, followed by aerobically incubated for 6 days without adding any extra microorganism. During the operation period, the indoor temperature was kept at 30 ± 2 °C, beakers were stirred by hand every 24 h with a neat glass rod. The substrates after 6 days aerobic pre-treatment was used as the feed during the anaerobic digestion process and stored in a −20 °C refrigerator until their use.

2.3.2. Semi-continuous anaerobic digestion process

Semi-continuous anaerobic digestion stage was carried out in a laboratory scale test to evaluate the production of CH_4. There are six reactors, corresponding to Pre-R1, Pre-R2, Pre-R3, R1, R2, R3, where R1, R2, R3 represent substrates without pre-treatment, with proportions of FW:SS were 1:0, 3:1, 1:1, respectively. Pre-R1, Pre-R2, Pre-R3 indicate that the substrates in these systems possess the same proportions of FW:SS as above, the only difference is that substrates have already gone through the aerobic pre-treatment. 250 mL Schott Duran bottle was used as the digester in this study. The organic loading rate (OLR) of each reactor was set to 4 g VS/L/d with the hydraulic reaction time (HRT) of 16 days. The working volume of the reactor was 100 mL, including 35 mL inoculum. Bottling operation once finished, the cap was inserted using silicone gel plug immediately and blown under a nitrogen atmosphere for 5 min to remove the oxygen (2 mL/s). After that, the reactor was maintained in a mesophilic temperature (37.5 °C) shaker at a speed of 150 rpm. The control reactor was filled with 35 mL inoculum and 65 mL distilled water to exclude the background CH_4 yield of the inoculum. 12.5 mL substrate was extracted and introduced every day by using a

syringe connected to the top lid. Before that, the headspace biogas was monitored to figure out its composition. All the data includes the average values. When the CH_4 production was steady, the anaerobic digestion experiment was stopped.

2.4. Analytical methods

2.4.1. Chemical analysis

The pH was measured by a pH meter (LeicipHS-3E, REX, China) equipped with temperature compensation. TS, VS, NH_4^+-N and TKN were analyzed by standard methods (APHA, 2005). Crude protein content was determined by multiplying Kjeldahl Nitrogen (KN) amount by the factor 6.25 [21]. Lipids were extracted with petroleum ether and quantified by measuring the mass loss of the residue after combustion [22]. The solute protein concentration and the solute carbohydrate concentration were determined by the Lowry method [23] and the phenol-sulfuric acid method [24], respectively. Elemental analyzer (Vario EL III, Elementar, Germany) was utilized for ultimate analysis adopting carbon-hydrogen-nitrogen model.

2 mL extract was extracted from each reactor every day and centrifuged at 13,500 rpm for 10 min and the supernatant was filtered with a 0.45 μm membrane filter (Millipore, USA) for the analysis of volatile fatty acids (VFAs), the rest was estimated for other parameters. For the detailed methods of sampling for VFAs, refer to Zhang et al. [25]. Gas chromatograph (GC-7900, TECHCOMP, Shanghai) with a thermal conductivity detector (TCD) and column packed TDX-01, was used to determine the volume of CH_4, CO_2 and N_2. Temperatures of injection port, column oven and detector were set to 100, 80, 100 °C, respectively. VFAs were also measured in the same GC-7900 equipped with a flame ionization detector (FID) coupled to a capillary column (GsBP-FFAP, 30 m × 0.25 mm × 0.25 μm). The carrier gas used is Argon at a flow rate of 44 mL/min and the split ratio was 1:20. The oven temperature was set to 80 °C and for one minute, then increased to 200 °C at 5 °C/min, which also lasted for one minute. The temperatures of the detector and injection port were at 260 and 250 °C, respectively.

2.4.2. FTIR analysis

The organic functional groups of the substrates were measured by FTIR spectrometry (Nicolet 6700 Flex, Thermo Fisher Scientific). Before analysis, all the substrates were dewatered to consistent weight at 105 °C. Solid was shredded into powder. In order to compare the intensity of the peaks, the ratio of all the samples and KBr was 1:200, the resolution is 4 cm^{-1}, 64 scans were collected, the wave number was 400–4000 cm^{-1}.

2.4.3. Microbial community characterization

Samples collected from systems on stable stage (66 days) were used for high-throughput sequencing analysis. Before analysis, the supernatant of samples was drained and then the remaining part was frozen at −20°C. The primers of (5′-TGYCAGCCGCCGCGGTAA-3′) and Arch958-RmodR (5′-YCCGGCGTTGAVTCCAATT-3′) were used for the polymerase chain reaction (PCR) amplification of V4-V5 *Archaea* (Shanghai Majorbio Bio-Pharm Technology Co. Ltd., China). PCR reactions were performed in triplicate 20 μL mixture containing 4 μL of 5 × FastPfu Buffer, 2 μL of 2.5 mM dNTPs, 0.8 μL of each primer (5 μM), 0.4 μL of FastPfu Polymerase, 0.2 μL of BSA, and 10 ng of template DNA under the following program: 95 °C for 3 min, 35 cycles at 95 °C for 30 s, 55 °C for 30 s, and 72 °C for 45 s, and a final extension at 72 °C for 10 min.

3. Results and discussion

3.1. Effects of simple pre-aeration on the semi-continuous anaerobic digestion process

3.1.1. Production of CH_4

The daily volumetric CH_4 production rate and CH_4 yield profiles are displayed in Fig. 1(a) and (b), including different ratios of FW:SS with and without pre-treatment. The experiment was conducted for a total number of 66 days. The CH_4 productivity and CH_4 yield were almost kept stable for this duration and they were distinctly different among the reactors with or without pre-aeration. During the start-up period (1–9 day), as the running time increases, CH_4 production of Pre-R2, R2, Pre-R3, R3 from zero climbed to 796.1, 407.8, 856.5, 433.5 mL/L·day, respectively. The reason why lower production of CH_4 in the primary stage is that, at start-up stage microbial physiological metabolic activity is low, thus leading to the volatility of the system. The total CH_4 yield and productivity in current study were lower than others, due to the short HRT (16 days) and only 35 mL inoculum was introduced into each reactor. In similar studies, the ratio of inoculum: substrate was 3 times higher than this [26,27]. After the start-up period, the systems is in stable state, at this stage, stable CH_4 productivity and CH_4 yield were obtained. That is, CH_4 productivity of Pre-R2 and Pre-R3 kept at 800 mL/L·day, while that of R2 and R3 were 450 mL/L·day. With the exception of pre-aeration treatment, Pre-R2 and R2 were operated under the same conditions, but, the CH_4 production was absolutely different. The CH_4 productivity of R2 was almost twice lower than that of Pre-R2. Similar result was existed in Pre-R3 and R3. It could be seen that pre-aeration is beneficial to the CH_4 production. Nevertheless, FW:SS ratio of feed substrate in R2 and R3 or Pre-R2 and Pre-R3 were different, but, the CH_4 production in R2 and R3 or Pre-R2 and Pre-R3 are basically the same, indicating that under the same OLR, ratios of FW:SS = 1:1 and 3:1

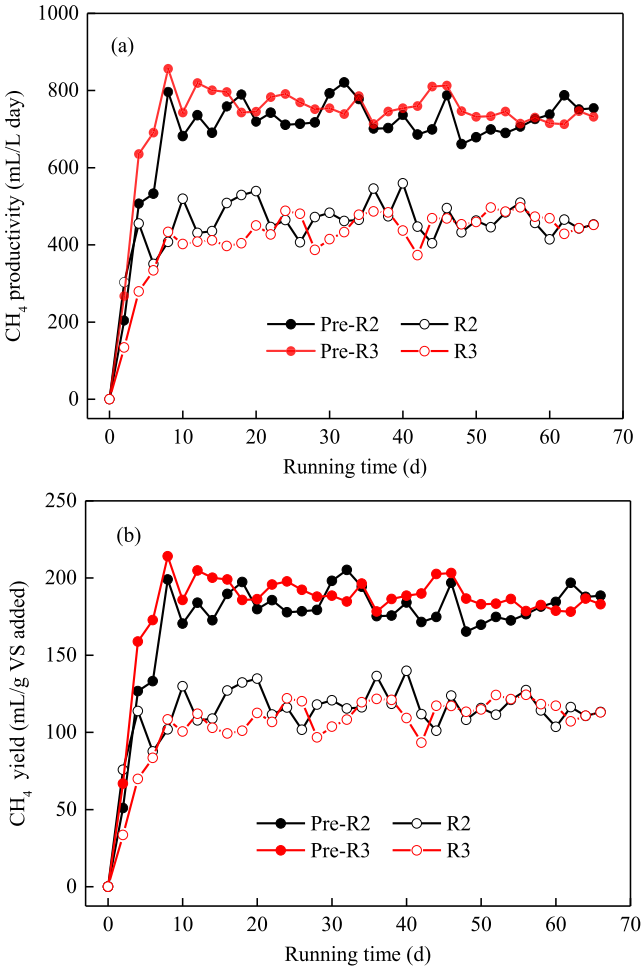


Fig. 1. Evaluation of the effects of pre-treatment on the biogas profiles during the semi-continuous anaerobic digestion process: volumetric CH_4 production rate (a); CH_4 yield based on the added VS (b). OLR and HRT were 4 g VS/L day and 16 days, respectively.

did not exhibit significant difference on the CH_4 yield.

Fig. 2(a) revealed that Pre-R1 and R1 did not show the predominant performance of CH_4 production when food waste was the only feed substrate. Fig. 2(b) and (c) illustrated the pH and VFAs of Pre-R1 and R1, VFAs accumulation exceed 7000 mg/L and the pH value was below 6.0 at the third day. The high VFAs concentration caused the inhibition of methanogenesis [28]. Lay et al. [29] assumed that the anaerobic digestion process would fail if pH was lower than 6.1. Combined with Fig. 1, it can be proved that co-digestion is an effective method for treatment of FW. Without pretreatment or mixing with other wastes, the food waste, alone, cannot run successfully at a high OLR. So, the Pre-R1 and R1 process was stopped when no more CH_4 was produced at 22 days. It is worth noting that though it was hard to carry out complete anaerobic digestion of only food waste, the gas production of the Pre-R1 is higher than that of the untreated system (R1), since it produced CH_4 until 22 days. It demonstrated that after pre-aeration process, the production of CH_4 could be improved even though the single anaerobic digestion of FW, i.e., aerobic pre-treatment in the process is very beneficial to generate more energy. In the anaerobic digestion process of MSW, Ni et al. [18] found accumulated biogas production can also be effectively improved with 8-days short-term pre-aeration.

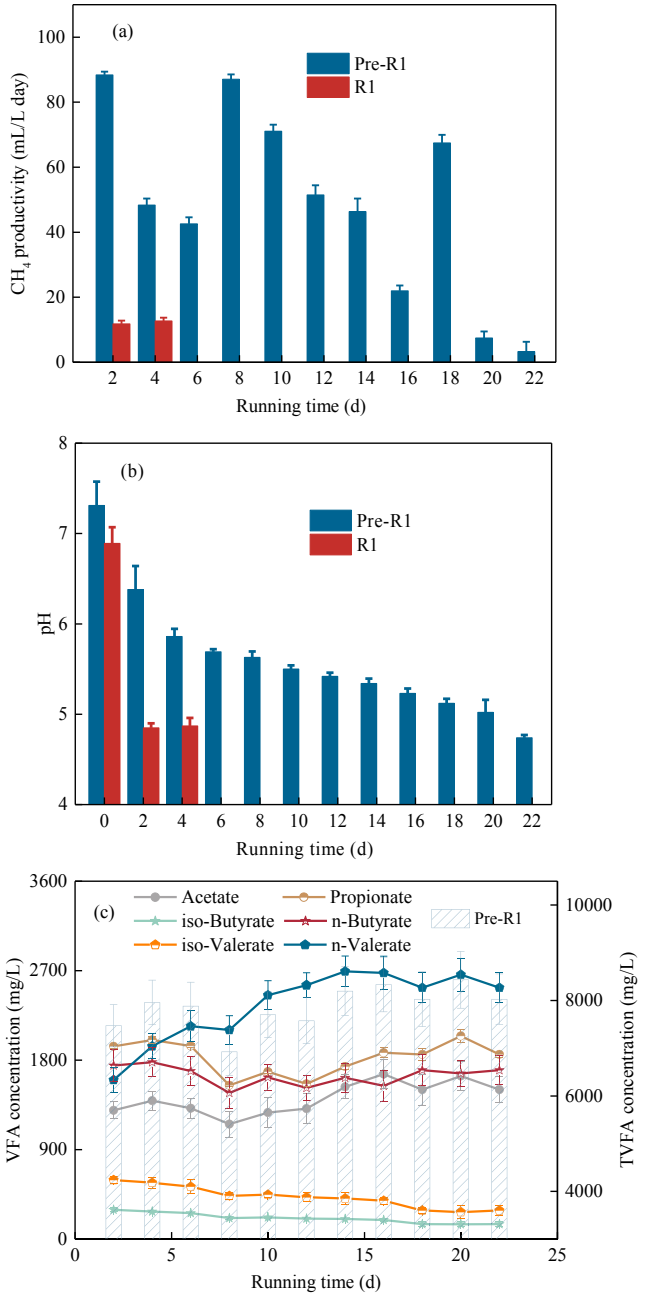


Fig. 2. Evaluation of the effects of pre-treatment on the biogas profiles during the semi-continuous anaerobic digestion process: CH_4 productivity of Pre-R1 and R1 (a); pH of Pre-R1 and R1 (b); TVFA and VFA concentrations of Pre-R1 (c), respectively.

Produced CH_4 content was, a reflection of the activities of hydrolyzing bacteria, acidogens and methanogens [30]. Fig. 3 shows higher CH_4 content in pretreated group in 66 days, which means pre-aeration increased the methanogenic activity. Xu et al. [31] reported that greater than 40% CH_4 content is required for the biogas to be technically or economically feasible. In this study, CH_4 content was maintained over 60%, representing the high quality of originated biogas.

From the Fig. 1, it displayed that the pre-treatment truly enhanced the yield of CH_4, since the CH_4 productivity was 60% higher for pre-treated array as compared to un-treated ones at the end of the operation period. In addition, co-digestion is a reliable approach to strengthen the performance of the anaerobic digestion of FW. The ratios (3:1 or 1:1) of FW:SS did not have a significant influence on the CH_4 yield or its productivity. It means with the appropriate ratio of FW:SS, the CH_4 productivity is independent of substrate composition under the same VS. Therefore, in this study, FW:SS = 3:1, with 6-days pre-aeration, is a preferred choice when considering the efficiency of food waste disposal and CH_4 production.

3.1.2. *The trend of pH and VFAs in the process of anaerobic digestion*

The pH value, TVFA concentration throughout the anaerobic digestion process and NH_4^+-N concentration after the pre-aeration are shown in Fig. 4. pH value is an important parameter to judge whether the reactor is operating normally. Fig. 4(a) showed high pH values above 7.6 were observed at the beginning of the anaerobic digestion process, and then, pH decreased in all runs due to the fast conversion to acids of substrates. The final pH maintained within the range of 7.05–7.40 on stable stage. Pre-R3 had the considerably higher pH (7.35), compared to Pre-R2 (7.15), R2 (7.08) and R3 (7.15), those pHs were desirable for anaerobic digestion process [32]. The decrease of pH 20 days at the start-up time of the reaction could be contributed by two reasons. One is the activity of methanogen was low at beginning, hence, the acids were not fully converted to methane. Another reason is the low buffer capacity of the substrate, which led to the accumulated acids were not able to be balanced in time. pH values are consistent with the trend of the TVFA. A high TVFA was observed in the first 20 days (shown in Fig. 4 (b)), but, with the rapid growth of methanogens, the accumulated acid was gradually utilized by methanogens. They were all below the inhibition concentrations. Concentrations of acetate, propionate, *iso-*

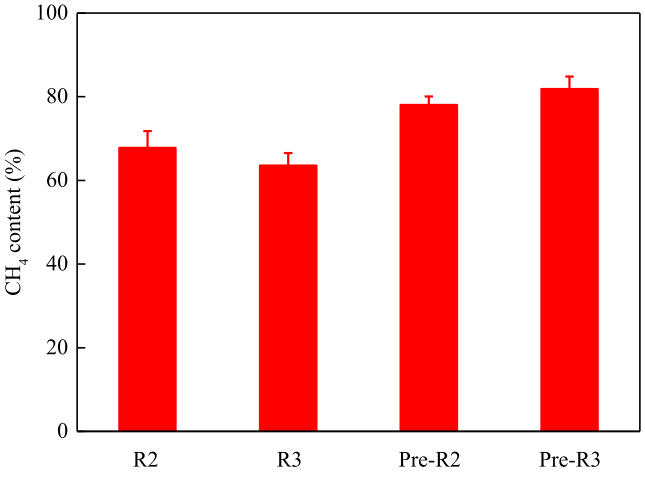


Fig. 3. CH_4 content in biogas of R2, Pre-R2, R3, Pre-R3 in 70 days.

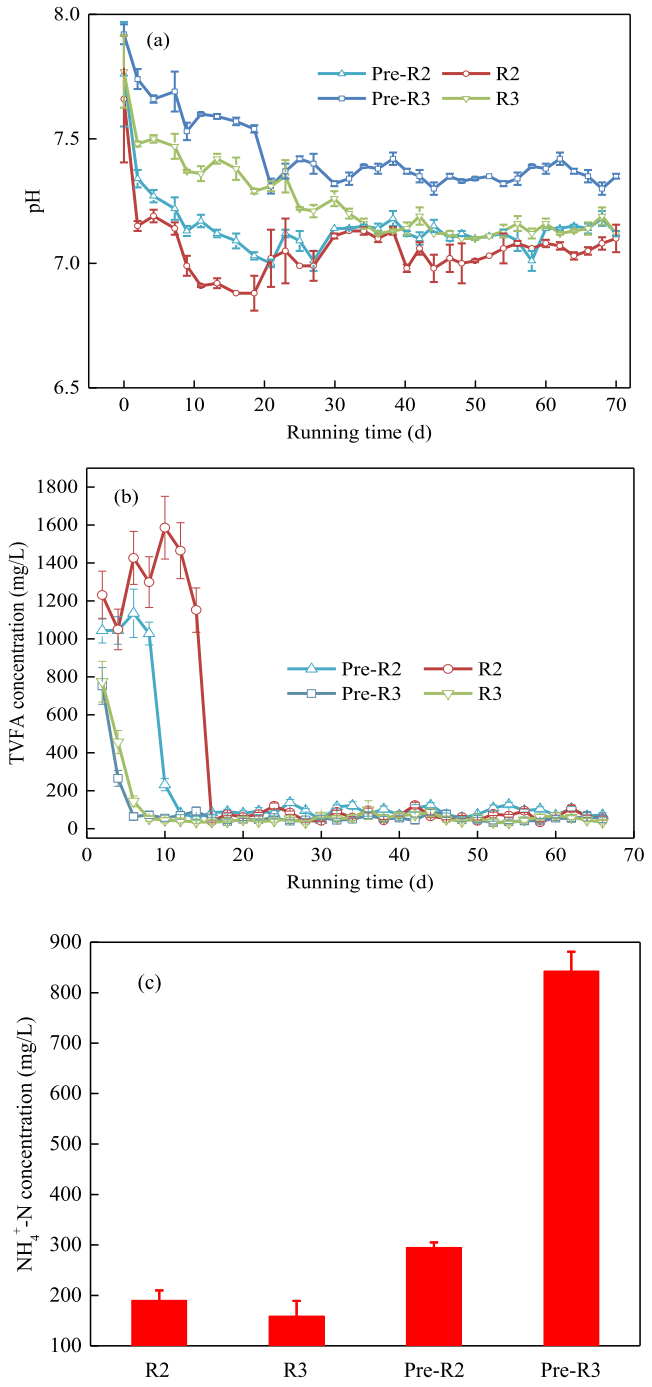


Fig. 4. Evaluation of the effects of pre-treatment on the pH (a); TVFA concentration profiles during the semi-continuous anaerobic digestion process (b); the NH_4^+-N concentration of feed substrates introduced into the systems (c).

Butyrate, *n*-Butyrate, *iso*-Valerate were also measured in Fig. 5. It is worth noting that *N*-Valerate was not detected all the time. All of them were at low values after start-up time. Fig. 4(c) shows the NH_4^+-N concentrations of feed substrates in anaerobic digestion process. After pre-aeration treatment, NH_4^+-N concentrations in the substrates were higher than non-pretreated arrays. NH_4^+-N concentrations were 190, 159, 295 and 843 mg/L for R2, R3, Pre-R2 and Pre-R3, respectively. Ammonium was mainly generated by protein degradation in substrate, which could be used to buffer the VFAs produced in the subsequent anaerobic digestion process and improve the CH_4 production. Aerobic pre-treatment greatly enhanced the degradation of protein in advance and released NH_4^+-N into bulk liquid, corresponding to the first higher pH values in Pre-R2 and Pre-R3. Rajagopal [33] reported that low NH_4^+-N concentration may cause low CH_4 yield, in accordance with the non-pre-aeration arrays which showed lower CH_4 productivity. However, it was suggested that when NH_4^+-N concentration reached to 1500 mg/L, an inhibition of methanogen activity would occur [34]. In this study, NH_4^+-N concentrations of systems were all below the 1500 mg/L, thus, pre-treatment series with high NH_4^+-N could dilute the generated VFAs and did not inhibit the CH_4 production. In previous studies, Ni et al. [18] demonstrated that a suitable pre-aeration duration was critical for the dry anaerobic digestion of MSW by reducing the organics and providing a buffer to neutralize the excessive produced VFAs.

These results, once again, suggested that the simple pre-aeration is an effective method for co-digestion of FW and SS. After 6-days aeration pre-treatment, higher pH and NH_4^+-N concentration of substrates were observed, which are favorable for methanogens. Thus dramatically creating effective conditions for the stabilization and CH_4 production in the mesophilic food waste anaerobic digestion process.

3.2. *The organic distribution of different substrates*

FW:SS = 3:1 is a preferred choice, therefore, this ratio was opted to determine the influence of pretreatment and non-pretreatment on the distribution of organic matter in the substrate. Protein and carbohydrate were the main components of substrate and played an important role in the growth of microorganisms in the system. However, the fermentation of microorganisms only use the solute organic components in the liquid phase, the content of solute proteins and carbohydrate in the fermentation system were critical to the efficiency of acid production, they are more easily solubilized, absorbed into the cells of microorganism and metabolized [35,36]. Solute protein and carbohydrate of substrates were detected in Fig. 6. The concentration of solute protein was 256 and 350 mg/L in the untreated and pretreated groups, respectively. The solute carbohydrate concentration was 198 and 667 mg/L, respectively. It was demonstrated that pretreatment promoted the following anaerobic digestion by converting macromolecular organic matter into available dissolved organic matter. High rates of organic matter degradation could be achieved by aerobic operation [37,38]. Considering that hydrolysis as the limited step in the single-phase anaerobic digestion process, pre-aeration promoted the CH_4 production by completing the partial hydrolysis stage in advance.

Further investigation by FTIR analysis was carried out after drying the pretreated and non-pretreated substrate, as shown in Fig. 7. The peak appeared in several wavelengths of the sample, which indicated the complexity of the mixture of FW and SS. At the same time, the peak position of untreated and pretreated samples was similar, which suggested that analogous functional groups existed in the untreated and pretreated reactors. The peak at wave number 1600–1700 cm^{-1} was related to N–H bending band of amide-I [39]. It was noted that the peak intensity at 1600–1700 cm^{-1} increased after aerobic pretreatment, indicating that aerobic pretreatment accelerated the degradation of proteins. The band at 1000–1190 cm^{-1} was the vibration of C–O, C–OH and C–C in carbohydrates, aromatic ethers and polysaccharides, respectively, which was related to the carbohydrate and aromatic ether substances [40]. After aerobic pretreatment, the peak weakened, which implied that aerobic pretreatment could promote the decomposition of carbohydrates into small molecules, thus reducing the content of carbohydrates in the substrate.

In general, some macromolecular organic substances in the substrate were broken down into small molecules before anaerobic digestion. The introduction of aerobic pretreatment fundamentally changed the structure of functional groups of substrates, thereby, promoting the biodegradability of the substrates [41]. Even under the same load in the anaerobic digestion step, it laid an excellent environment for methanogens and remarkably amended the CH_4 growth rate, compared with the group without pretreatment.

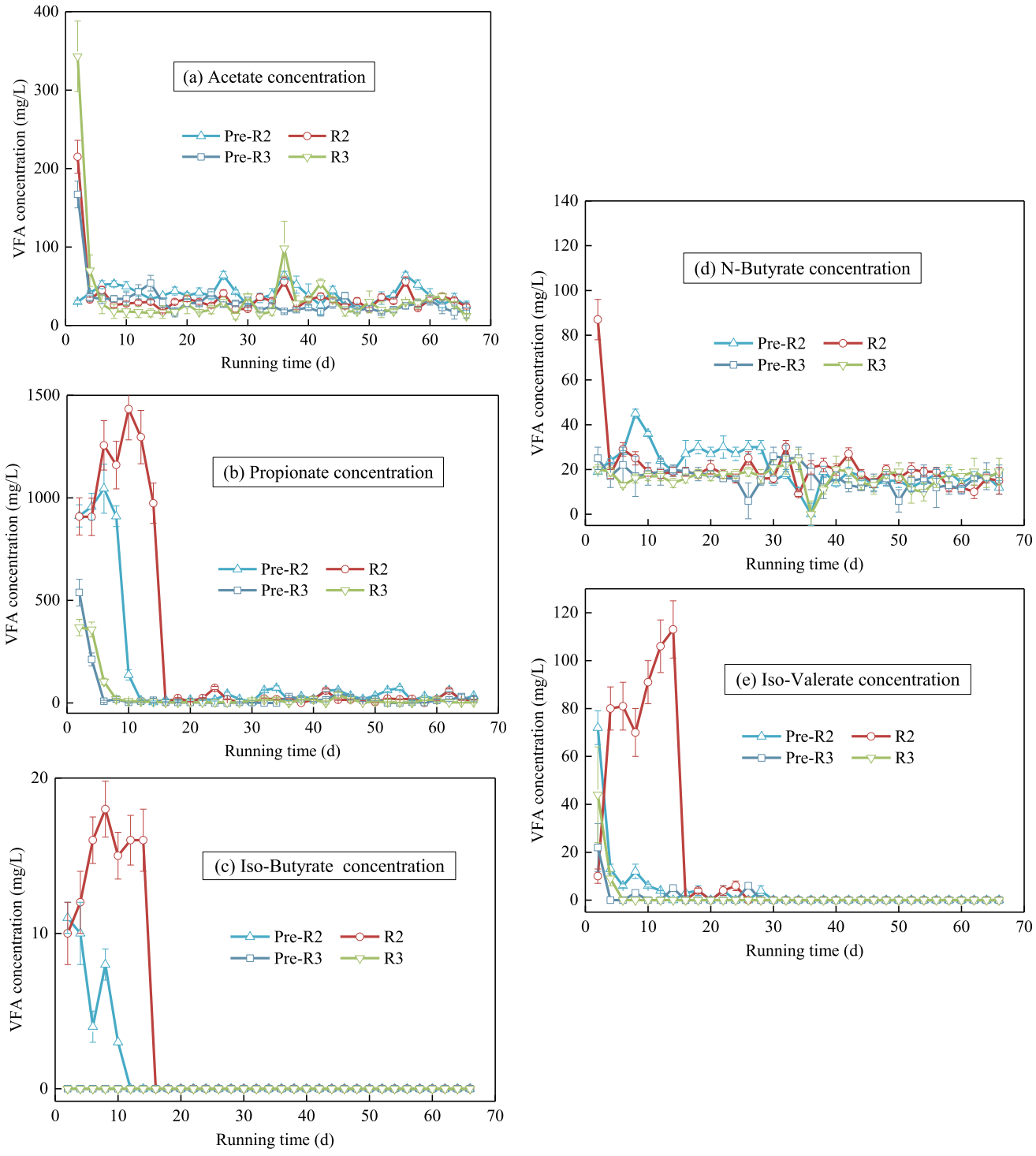


Fig. 5. Concentration of Acetate (a); Propionate (b); Iso-Butyrate (c); *N*-Butyrate (d); Iso-Valerate (e) during anaerobic digestion process.

3.3. Evolution of function microorganisms

In order to understand the influence of pretreatment and FW:SS ratio on methanogenic community in the operation system, 16S RNA was used to measure the community structure difference of *Archaea* in different reactors at the stable stage. The *Archaea* community structure of systems at genus level was presented in Fig. 8. *Methanosarcina* are found as the major methanogen, with a high abundance of more than 60%. In addition, there are *Methanobacterium*, *Methanosaeta*, *Methanofollis*, *Methanomassiliicoccus*, *Methanospirillum*, etc. Total relative abundance of these methanogenic *Archaea* is over 98%. Among them, *Methanosarcina* can use methyl compounds such as acetate, H_2/CO_2, methanol and methylamine to form CH_4. Therefore, it has strong adaptability and is a mixed methanogen, which is regarded as the dominant strain in anaerobic digestion process. In a stable anaerobic digestion reactor, *Methanosarcina* generally occupies the main position, which can be up to more than 70% [42,43]. *Methanosaeta* is a specific acetotrophic methanogen while *Methanobacterium* is a hydrogen-utilizing group. Moreover, *Methanospirillum* is a type of facultative acetate-utilizing group methanogen, *Methanofollis* only use hydrogen, carbon dioxide and formate to produce CH_4 while *Methanomassiliicoccus* metabolizes mixed nutrient (methyl and hydrogen) for CH_4 production.

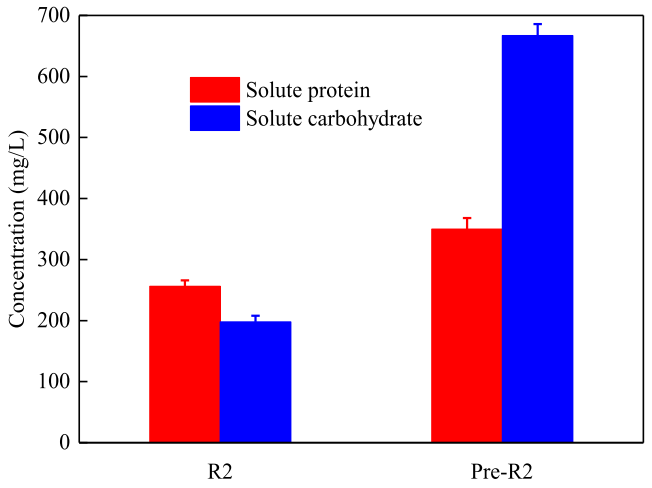


Fig. 6. Solute protein and solute carbohydrate concentration of substrates in Pre-R2 and R2, after 6-days pre-aeration and without pre-aeration, respectively.

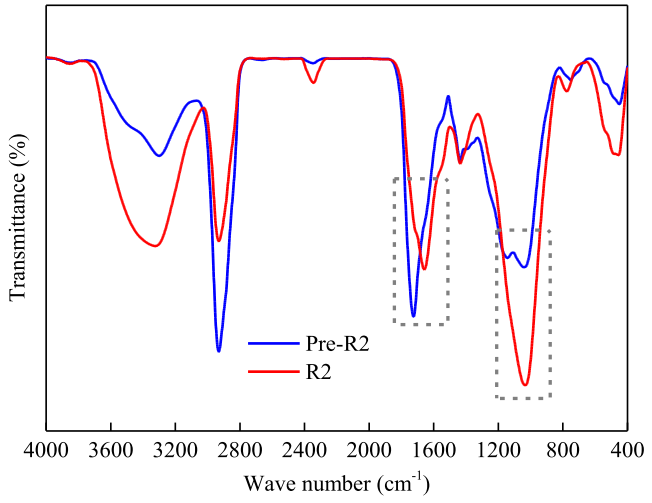


Fig. 7. FTIR spectra of dried substrates in Pre-R2 and R2 after 6 days pre-aeration and without pre-aeration, respectively.

The distribution of relative abundance of *Archaea* has changed greatly under the long-term operation of pretreated and untreated substrates. As can be seen from the Fig. 8, in the system without pretreatment, the abundance of *Methanosarcina* in R2 was lower than that of pretreatment, and the abundance of *Methanosarcina* in Pre-R3 and Pre-R2 was 93.10% and 95.98%, respectively. While the abundance in R2 was only 62.22%, and *Methanobacterium* accounted for 32.57% in R2. It was found that *Methanosarcina* was mainly present in the stable operation, while *Methanobcterium* accounted for a small proportion. The less dominance of *Methanosarcina* resulted in lower CH_4 production in untreated group than that in pretreated group. This result was consistent with Sang's study [44], high CH_4 production was observed in the intermittently aerated reactor, where *Methanosarcinales* as acetate-utilizing methanogens, were dominant *Archaea*. Compared with different FW:SS ratios, the abundance of *Methanosarcina* in Pre-R3 was lower than that in Pre-R2, but it was noted that compared with Pre-R2, *Methanosaeta* in Pre-R3 was 3.02%, thereby, making up for the deficiency of *Methanosarcina*, as acetate-utilizing methanogen.

To sum up, after aerobic pretreatment, the substrate entered the anaerobic system, which would change the abundance of different methanogens in the system. Methanogens in aerobic pretreatment group were mainly *Methanosarcina*, while methanogens in non-pretreatment group were mainly *Methanosarcina* and *Methanobacterium*. As a result, the main pathway in pre-aerated arrays was acetotrophy for methanogenesis and methanogenic performance for non-pre-treatment group was weaker than that of pretreatment group.

4. Conclusions

The effect of 6-days aerobic pretreatment on co-digestion of FW and SS was studied. The results showed that aerobic pretreatment could increase the concentration of ammonia nitrogen in the substrate, which effectively buffer the acid produced and promote the CH_4 production. After aerobic pretreatment, the maximum CH_4 production capacity of the system was 856 mL/L day, and the VFAs concentration remained at low level after the successful start-up of the system. Furthermore, after the aerobic pretreatment, macromolecular organic matter (protein and carbohydrate) could be degraded into soluble protein and carbohydrate for utilization by fermentation microorganisms. At the same time, high-throughput sequencing showed that with the long-term operation of the reaction, the microbial community in the system altered due to the pre-aeration. In the aerobic pretreatment group, the *methanosarcina* was the main microorganism, while the non-pretreatment group contained 32.57% *methanobacteria*. These results demonstrated that a short pre-

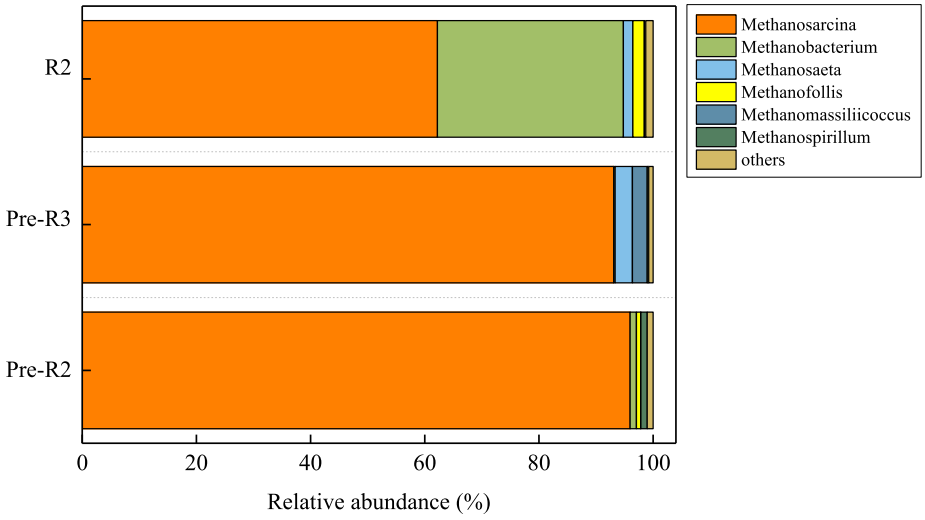


Fig. 8. High-throughput 16S rRNA gene pyrosequencing analyses of the archaeal community structure at genus level.

treatment intensified the CH_4 production by deforming the feed substrate of anaerobic digestion process. Aerobic pretreatment was considered as a reliable method to improve the CH_4 production performance of FW and SS co-digestion. The results of this research are beneficial to promote the CH_4 production from anaerobic digestion of FW and can facilitate the practical application of the projects in the future.

CRediT authorship contribution statement

Lijie Cheng: Data curation, Formal analysis, Methodology, Writing - original draft. **Ningbo Gao:** Conceptualization, Funding acquisition, Methodology, Supervision, Writing - original draft. **Cui Quan:** Project administration, Resources. **Hua Chu:** Resources, Writing - review & editing. **Guojuan Wang:** Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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